

1. List 5 different types of cyber security threats you might encounter while using popular apps like Instagram, WhatsApp, or Zomato, and give a real-life example for each.

No	Cybersecurity Threat	What it means	Real-life example
1	Phishing	Fake messages, emails or links trick users into revealing passwords or personal information.	An Instagram user receives a message saying, "Your account will be deleted. Click this link to verify". The link leads to a fake Instagram login page that steals their password.
2	Account Hacking	Attackers obtain your password and gain unauthorized access to your account.	Someone uses a password leaked from another website to log into user's Instagram account and changes password and recovery email.
3	Malware	Malicious software can be hidden in files, apps or links and may steal data or damage a device.	A WhatsApp user receives an APK file claiming to be a new version of WhatsApp. After installing it malware accesses their messages and other phone data.

4	Fake Apps and Websites	Criminals create fake versions of popular apps or websites to steal information or money.	A users searches for a Zomato offer online and visits a fake website offering a 90% discount. They enter their login or payment details which are captured by the attacker.
5	Social Engineering	Attackers manipulate people psychologically into sharing information or sending money.	A scammer contacts a WhatsApp user pretending to be a friend and says, "I'm stuck and urgently I need 5000 rupees". The victim sends money without verifying the person's identity.

2. Simulate a phishing attack scenario by designing a fake login page for an app you use (for example, a mock Instagram or Flipkart login screen in HTML). Explain how a user could be tricked and what signs would help spot the threat.

Instagram

Sign in to continue

Log In

⚠ PHISHING SIMULATION — NO DATA IS COLLECTED

- Above I stimulate a fake Instagram login page for a phishing.

How a user a could be tricked

- An attacker could send someone a message such as: “Your account will be suspended. Verify your account immediately”.
- The message could contain a link leading to a fake login page that visually looks like Instagram. A person who trusts the message might enter their username and password without checking the website carefully.

Signs to identify phishing page

- Suspicious URL – the address may contain a misspelled or unrelated domain rather than the legitimate website.
- Urgent language – messages such as “Your account will be deleted in 10 minutes” pressure users into acting quickly.
- Unexpected login request – be suspicious if a link unexpectedly asks you to log in.
- Spelling/grammar mistakes – poorly written messages can indicate phishing.
- Requests for sensitive information – never provide passwords, OTPs or other sensitive information through suspicious links.
- Missing security indicators – HTTPS alone does not provide a website is legitimate because domain also must also be checked.

- 3. Research and write a short report (150-200 words) on a recent cyber-attack that affected a major Indian tech platform (such as Paytm, IRCTC, or PhonePe). Describe the type of threat, how it was carried out, and what was done to fix it.**

Cyber-attack on IRCTC

- IRCTC one of India’s largest online ticketing platforms has faced repeated cyber threats particularly during high-demand tatkal booking periods. A major concern has been automated bot attacks and fraudulent account activity where attackers use bots and large number of suspicious account to overload booking systems and gain unfair access to tickets. In 2025, IRCTC deactivated around 3.03 crore suspicious user accounts and reported hundreds of cybercrime complaints involving suspicious PNRs.

- To counter these threats, IRCTC introduced AI/ML-based monitoring and anti-bot technology to identify unusual booking patterns and block malicious traffic. Its anti-bot system reportedly mitigated about 64% of malicious traffic. IRCTC also uses a Content Delivery Network (CDN), Web Application Firewall Intrusion Prevention Systems (IPS) and DDos-mitigation measures to protect its infrastructure. Suspicious email domains and user accounts are blocked or deactivated while genuine users can undergo revalidation to regain access.
 - The IRCTC shows how automated attacks can disrupt online services and why continuous monitoring, bot detection and layered cybersecurity defences are essential.
4. Pick any one cyber security threat (like ransomware, phishing, or DDoS) and create a simple infographic or poster (hand-drawn or digital) that explains how it works, who is at risk, and how to prevent it.

